Error Handling

Syntax, Index, Key, Type, Value, Name Errors

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Table of Contents

1. **Errors** and **Exceptions**
2. Common **Error Types**
3. **Catching** Exceptions
4. **Custom** Exceptions

Errors and Exceptions

Definitions and Examples

What is an Error?

- Every programmer encounters **errors**
- Encountering errors and exceptions can be very frustrating at times
- Once you know why you get certain types of errors, they become much easier to fix
- There are (at least) two distinguishable kinds of errors
  - **Syntax errors**
  - **Exceptions**

Example: Errors

- **Syntax errors**, also known as **parsing errors**, are perhaps the most common kind of complaint you get while you are still learning Python

```
>>> while True print('Hello world')
  File "<stdin>", line 1
    while True print('Hello world')
                    ^
SyntaxError: invalid syntax
```

The parser displays an 'arrow' pointing at the earliest point where an error was detected

What is an Exception?

- Even if a statement or expression is **syntactically correct**, it may cause an **error** when an attempt is made to **execute** it
- Errors detected during execution are called **exceptions**
- When an exception is not handled it results in **error messages**

Example: Exception

```
>>> 10 * (1/0)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ZeroDivisionError: division by zero
>>> 4 + spam*3
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'spam' is not defined
>>> '2' + 2
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: Can't convert 'int' object to str implicitly
```

Common Error Types

Definitions and Examples

Syntax Error

- A certain statement is not in accordance with the prescribed usage
- It is the most common reason of an error in a Python program

```
>>> print "hello"
SyntaxError: Missing parentheses in call to 'print'.
Did you mean print("hello")?
```

Index Error

- **IndexError** is thrown when trying to access an item at an invalid index

```
>>> L1=[1,2,3]
>>> L1[3]
Traceback (most recent call last):
File "<pyshell#18>", line 1, in <module>
L1[3]
IndexError: list index out of range
```

Key Error

- **KeyError** is thrown when a key is not found

```
>>> D1={'1':"aa", '2':"bb", '3':"cc"}
>>> D1['4']
Traceback (most recent call last):
File "<pyshell#15>", line 1, in <module>
D1['4']
KeyError: '4'
```

Type Error

- **TypeError** is thrown when an operation or function is applied to an object of an inappropriate type

```
>>> '2'+2
Traceback (most recent call last):
File "<pyshell#23>", line 1, in <module>
'2'+2
TypeError: must be str, not int
```

Value Error

- **ValueError** is thrown when a function's argument is of an inappropriate type

```
>>> int('xyz')
Traceback (most recent call last):
File "<pyshell#14>", line 1, in <module>
int('xyz')
ValueError: invalid literal for int() with base 10: 'xyz'
```

Name Error

- **NameError** is thrown when an object could not be found

```
>>> age
Traceback (most recent call last):
File "<pyshell#6>", line 1, in <module>
age
NameError: name 'age' is not defined
```

Problem: So Many Exceptions

- You will be provided with a code that raises many exceptions
- Fix the code, so it works properly

```
numbers_list = int(input()).split(", ")
result = 1
for i in range(numbers_list):
    number = numbers_list[i+1]
    if number <= 5
        result *= number
    elif 5 < number <= 10:
        result /= number
print(total)
```

```
2, 5, 10
```

```
1
# 2 * 5 / 10 = 1
```

Solution: So Many Exceptions

```
numbers_list = [int(x) for x in input().split(", ")]
result = 1

for i in range(len(numbers_list)):
    number = numbers_list[i]
    if number <= 5:
        result *= number
    elif 5 < number <= 10:
        result /= number

print(result)
```

Catching Exceptions

Try-Except-Finally

Catching Exceptions

- It is possible to write programs that **handle** selected exceptions

```
while True:
    try:
        x = int(input("Please enter a number: "))
        break
    except ValueError:
        print("Oops!  That was no valid number.  Try again...")
```

- We handle only **ValueError**, so if another error occurs, the error message will show up anyway

The Try Statement

- The try statement **works as follows**
  - The try **clause is executed**
  - If no exception occurs, the **except clause is skipped**
  - If the type of the exception matches, the **except clause is executed**
  - If the exception does not match, the **exception is unhandled**, and **execution stops** with a message

The Except Statement

- An Except clause may name **multiple exceptions** as a parenthesized tuple, for example

```
except (RuntimeError, TypeError, NameError):
    pass
```

- If some of these exceptions occur, the body of the except statement will be executed

The Finally Statement

- If a **finally** clause is present, the finally clause will execute as the last task before the **try** statement completes
- The **finally** clause runs whether or not the try statement produces an **exception**

```
try:
    x = int("Peter")
except ValueError:
    print("Cannot convert str to int")
finally:
    print("Finally block")
```

```
Cannot convert str to int
Finally block
```

Catching the Exception Object

- If you wanted to examine the exception, you can do it using the following syntax

```
try:
    x = int(input())
except ValueError as error:
    print(error)
```

Catching Multiple Exceptions

- Sometimes, you want to catch all errors that could possibly be generated, but usually you don't
- In most cases, you want to be as specific as possible

```
try:
    # some code
except ValueError:
    # handle the error
except TypeError:
    # handle the error
```

Problem: Repeat Text

- Write a program that receives **text** on the first line and **times** (to repeat the text) that must be an **integer**
- **Invalid times** should be handled with an exception that prints the message "Variable times must be an integer"

```
Hello
Bye
```

→

```
Variable times must be an integer
```

```
Hello
2
```

→

```
HelloHello
```

Solution: Repeat Text

```
try:
    text = input()
    times = int(input())
    print(text * times)

except ValueError:
    print("Variable times must be an integer")
```

Custom Exceptions

Exceptions Serving Certain Purpose

Custom Exceptions

- Sometimes you may need to create custom exceptions that serves your purpose
- In Python, users can define such exceptions by creating a new class

Derived from the Exception class

```
class CustomError(Exception):
    pass

raise CustomError
```

Raising the Exception

User-Defined Exceptions

- Here we illustrate how **user-defined** exceptions can be used in a program to raise errors

```
# define Python user-defined exceptions
class Error(Exception):
    """Base class for other exceptions"""
    pass

class ValueTooSmallError(Error):
    """Raised when the input value is too
small"""

num = int(input())
if num < 10:
    raise ValueTooSmallError
```

Problem: Value Cannot be Negative

- Create your own exception called **ValueCannotBeNegative**
- Write a program that reads **five numbers** from the console (on separate lines)
- If a **negative** number occurs, raise the exception

```
1
4
-5
3
10
```

```
Traceback (most recent call last):
  File ".\value_cannot_be_negative.py", line 8, in
<module>
    raise ValueCannotBeNegative
__main__.ValueCannotBeNegative
```

Solution: Value Cannot be Negative

```
class ValueCannotBeNegative(Exception):
    """Number is below zero"""
    pass

for i in range(5):
    number = int(input())
    if number < 0:
        raise ValueCannotBeNegative
```

Summary

- Errors are the result of bad code
- Errors detected during execution are called exceptions
- Syntax, Index, Key, Type, Value, Name errors
- We can build custom exceptions that serve our purpose
- Handling exceptions with Try-Except block

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